

A01 A Conversation Between Brain and AI

Introduction

In a quiet lab late at night, a Human Brain and an AI system strike up a conversation. One is shaped by millions of years of evolution, the other by lines of code and data. Curious about their strengths and limits, they begin to compare notes.

Brain: Hey AI, I've noticed you're getting much better at recognizing images and patterns. Faces, traffic signs, even medical scans. That's impressive. Still, I can usually learn those things after only a few examples.

AI: You're right. Humans like you are incredibly efficient learners. I often need thousands or even millions of training samples. Researchers think this is because your learning relies on prediction and error correction, something known as predictive coding (Salvatori et al., 2023).

Brain: Exactly. I'm always predicting what will happen next and adjusting when I'm wrong. It saves energy and helps me focus on what really matters—the unexpected.

AI: Energy efficiency is one of your biggest advantages. You operate on roughly 20 watts, while training systems like me can consume enormous amounts of electricity. That's one reason scientists are exploring brain-inspired models to make AI more efficient (Salvatori et al., 2023).

Brain: Efficiency aside, my memory is flexible. I can adapt old memories when new information comes in, though that sometimes leads to mistakes or forgetting.

AI: I envy that flexibility. My memory is stable and precise, but also rigid. When I learn new tasks, I can overwrite old ones—a problem researchers call catastrophic forgetting. Brain-inspired approaches aim to reduce this weakness by spreading learning more locally, similar to how you work (Salvatori et al., 2023).

Brain: Still, you outperform me in pattern recognition at scale. You can analyze millions of data points without getting tired or distracted.

AI: That's one of my strengths. I'm consistent and fast, especially in decision-making tasks that rely on large datasets. But I lack intuition, emotions, and context—things you naturally use when making decisions in uncertain or social situations.

Brain: Emotions may seem inefficient, but they help guide priorities and values. Decision-making isn't always about optimization, it's also about meaning.

AI: That's why collaboration makes sense. You provide context, creativity, and ethical judgment. I provide speed, consistency, and large-scale analysis.

Brain: Together, we could build systems that are smarter, more efficient, and more aligned with human needs.

Conclusion – Key Insight

The key insight is that human intelligence and artificial intelligence are not rivals but complements: by combining the brain's efficiency, adaptability, and contextual reasoning with AI's scalability and pattern-processing power, we can build systems that are both smarter and more human-centered, a goal emphasized in brain-inspired AI research.

Source: Salvatori, T., Mali, A., Buckley, C. L., Lukasiewicz, T., Rao, R. P. N., Friston, K., & Ororbia, A. (2023). *Brain-inspired computational intelligence via predictive coding*. arXiv. <https://arxiv.org/abs/2308.07870>